

ABEER NASIR CHAUDHRY

Signal Processing/Algorithm Development Engineer



EDUCATION

Master of Science in Electrical Engineering

American University of Sharjah, UAE

Specialization: Radar Signal Processing, Deep Learning

Awards: Int'l Graduate Scholarship

September 2019 - June 2021

CGPA: 3.78 / 4.00

Bachelor of Engineering in Avionics Engineering

National University of Sciences and Technology (NUST), Pakistan

Awards: University Gold Medal, Best Final Year Project, Best Avionics System Design, Commandants Honors

October 2014 - February 2019

CGPA: 3.32 / 4.00

PROFESSIONAL EXPERIENCE

Digital Signal Processing Simulation Engineer

National Aerospace Science and Technology Park (NASTP)

December 2021 – Current

- Developed a novel solvability-region-aware root-dynamics method for selecting among analytical TDoA candidate solutions. A transition-aware regularization is added for samples near the leading-coefficient degeneracy of the quadratic. It uses the finite linear-limit root at transition entry and a local continuity recovery rule until the raw selector re-acquires a stable branch
- Designed and implemented a Quadratically Constrained Least Squares (QCLS)-based Time Difference of Arrival (TDoA) localization algorithm, leveraging iterative bias correction, EVD, Tikhonov Regularization and nonlinear optimization to enhance estimation accuracy
- Modeled, implemented and optimized a two-step grid search algorithm for Frequency Difference of Arrival (FDoA) based localization using Quasi-Newton optimization with RMSE of 200 meters at a maximum range of 400 km and computation in under 300 ms.
- Dynamically computed and integrated Geometric Dilution of Precision (GDoP) analysis to optimize sensor network configurations and improve positioning accuracy in localization and tracking systems for passive geolocation
- Developed a multi-lateration and sensor fusion algorithm for distributed asynchronous passive sensors with accuracy ranging between 20 meters and 1050 meters at a 400 km range
- Comprehensive stand-alone radar simulator design and development for SIL and HIL testing of RF front-end and signal processing chain for passive geolocation systems. This comprehensive simulator was completed in under 11 months
- Integrated RF channel and propagation medium modeling (Rician, Rayleigh, Multipath, AWGN, TIREM, ITU Stds.) with the simulator
- Comm and RADAR waveform synthesis and analysis (FMCW, CW, pulse, m-ary PSK, FSK, QAM etc.)
- Digital filter design, time-invariant wavelet denoising for signal reconstruction, wavelet MRA. By implementing these techniques for various OTA signals, the SNR was improved by up to 30% leading to a timing accuracy of less than 100 nanoseconds
- Resource constrained algorithm development for signal feature extraction

Graduate Research/Teaching Assistant

American University of Sharjah, UAE

September 2019 – June 2021

- Frequency Diverse Array (FDA) contemporary research review and identification of lapse in fundamental technical assumptions while considering the applications as a primary radar transceiver with Equivalent Transmit Beamforming.
- Study and successful real-time rectification of effect of mutual coupling in arrays and its mitigation via pre-distortion of the jointly estimated mutual coupling coefficients
- Finite-Difference Time Domain (FDTD) simulations of Frequency Diverse Arrays to study the effect of mutual coupling using HFSS with closed boundary conditions while neglecting the Pattern Multiplication effect
- Responsible for research work, paper reviews and summaries
- Managed over 5 different labs for undergraduate students, grading papers for 8 courses and conducted seminars/workshops

Avionics Design Engineer

Pakistan Aeronautical Complex, Pakistan

April 2019 – September 2019

- Member of the radar development team, responsible for radar feasibility studies and performance simulations for AESA radar

ACHIEVEMENTS

- Marie-Sklodowska Curie Actions Doctoral Networks (MSCA-DN) Fellowship Awardee (*Project INTEGRATE*, 2023)
- University International Postgraduate Award (UIPA) Scholarship, University of New South Wales (UNSW), Australia (2023)
- International Graduate Student Scholarship (2019-2021)
- Rectors' Gold Medal for the Best Final Year Project in Avionics Engineering (2019)
- Awarded Best Avionics Suite Design (2019)
- Awarded Best Community Service (2019)
- Academic Mentor College of Aeronautical Engineering, NUST (2017-2019)

SKILLS

Programming Languages: MATLAB (*Advanced*), Python (*Advanced*), SQL (*Intermediate*), LabView (*Intermediate*), Verilog (*Basic*)

Frameworks: OpenCV, TensorFlow, Keras, Sci-py, NumPy, Seaborn, Matplotlib, Pandas, XGBoost

Project Management and Version Control: Notion, Git

Engineering Design Software: HFSS (*Advanced*), MultiSim (*Advanced*), Advanced Design System (ADS) (*Advanced*), Vivado EDA (*Basic*)

Other Software/Tools: Microsoft Power BI (*Advanced*), Latex (*Advanced*), Microsoft Office (*Advanced*)

Soft Skills: Leadership, Resourceful, Meta-learning, Teamwork, Mentoring, Adaptability, Time Management, Analytical, Self-Motivated,

Languages: Native or Bilingual Proficiency in English and Urdu, Basic Proficiency in German and Chinese

ACADEMIC PUBLICATIONS

- A. N. Chaudhry, H. S. Mir and K. T. Wong, "Effect of Target Motion on Equivalent Transmit Beamforming for the Frequency Diverse Array", in *IEEE Transactions on Aerospace and Electronic Systems*, doi: 10.1109/TAES.2023.3282192.
- A. N Chaudhry, S Liaquat and M Khadim, " κ : A Geometry-Quality Metric Complementary to GDoP for Closed-Form TDoA Multilateration", Under Review in *IEEE Transactions on Aerospace and Electronic Systems*
Preprint available at: <https://arxiv.org/abs/2606.14372>
- M Khadim, S Liaquat and A. N Chaudhry "TokenDSP: Training-Free Prompt Compression via Spectral Residual and Information-Theoretic Analysis of Token Embeddings", Under Review in *IEEE Transactions on Audio, Speech and Language Processing*

PROJECTS

- **Strava Data Analysis and Visualization Project: Building an Interactive Health Dashboard (Data Analysis):** Exploratory Data Analysis (EDA) and data modeling using Python and Power BI. This will be achieved by analyzing my personal Strava workout data. The project will involve segregating thousands of workout files. The end goal is to create an interactive health dashboard similar to Apple and Samsung Health. **Platform:** Python, XGBoost, Power BI, seaborn, pandas (*in progress*)
- **Detecting Parkinson's Disease (Data Science, Machine Learning):** Parkinson's is a neurodegenerative disease that has no cure. As the world moves towards better healthcare, data is being leveraged to provide proactive healthcare. In this project the UCI ML Parkinson's dataset is used and an XGBoost classifier is built to accurately detect the presence of this disease in an individual. **Platform:** Python, XGBoost, sci-kit learn, pandas (*May 2024*)
- **Uber Data Analysis (Data analytics, visualization):** This is a data visualization project with ggplot2 using python and its libraries to analyze various parameters like trips by the hours in a day and trips during months in a year. The Uber Pickups in New York City dataset is used to create visualizations for different time-frames of the year. **Platform:** Python (*June 2024*)
- **Practical Considerations in Frequency Diverse Array Radar Signal Processing (Master's Thesis):** The influence of target motion on equivalent transmit beamforming is elaborated upon, and a metric is constructed to quantify the adverse consequences of a non-stationary target on system detection capabilities. Mutual coupling coefficients are jointly estimated with the target parameters to pre-distort the array to compensate for and eliminate mutual coupling among the antenna elements. **Platform:** MATLAB (*May 2021*)
- **Real-Time Automatic Facial and Emotion Recognition using Deep Learning (ConvNets, Transfer Learning, Computer Vision):** A deep-learning model that captures the faces via webcam and predicts and recognizes the facial emotions based on the Facial Action Coding System (FACS). Metrics benchmarked against various ILSCRC winners. **Platform:** Jupyter Notebooks, keras, OpenCV (*Oct 2019*)
- **UWB Radar Cross-section Reduction in a Compact Anti-podal Vivaldi Antenna (High frequency structure closed boundary design and analysis):** Employed object shaping techniques to reduce the RCS of an Antipodal Vivaldi antenna having highly directional beampattern with abundant use in airborne radar systems (AESA radar). **Platform:** HFSS (*Nov 2019*)
- **Maximum Likelihood Estimation Assisted Mitigation of the Effect of Noise Uncertainty in the State-of-Charge Estimation of Li-ion Battery Cells (Linear Estimation, Bayes' Theorem, Kalman Filter):** An efficient approach is devised and simulated to exhibit improved SOC estimation after estimating the noise uncertainty using MLE algorithm in conjunction with EKF. **Platform:** MATLAB. (*Oct 2020*)
- **Design, Analysis and Realization of a 7th Order Stepped Impedance LPF and its Comparison with Other Realizations (Microwave circuit analysis, multiport networks):** Stepped-Impedance architecture for a Maximally flat 7th order filter is designed and compared with O.C Stub Impedance design. Performance comparison of the EM simulation of the two architectures is shown. **Platform:** ADS (*March 2021*)
- **Real-Time Object Segmentation in Images using "PointRend" Segmentation Architecture (Computer Vision, Object Segmentation):** The state-of-the-art neural network architecture "PointRend" is implemented over the Coco dataset for object segmentation. The performance of different modes and models, *resnet101* and *resnet50* is compared. **Platform:** Jupyter Notebooks, TensorFlow (*Oct 2021*)

INTERNSHIPS

Pakistan International Airlines (PIA) – Engineering Wing. Avionics overhaul and repair management	September 2017 – October 2017
Pakistan Aeronautical Complex (PAC) – Avionics Production Factory. Design, Development and Deletion Group	March 2017 – April 2017
Huawei, Qatar – Communications Engineering Department	March 2016 – April 2016

MEMBERSHIPS

- Pakistan Engineering Council, Registration No. AERO/2597
- Young Member Royal Aeronautical Society (YRAeS)
- Volunteer Acts of Kindness (AoK), Pakistan

HOBBIES/EXTRA-CURRICULAR

- Drone photo/videography
- Traveling
- Endurance sports (cycling, running, swimming)